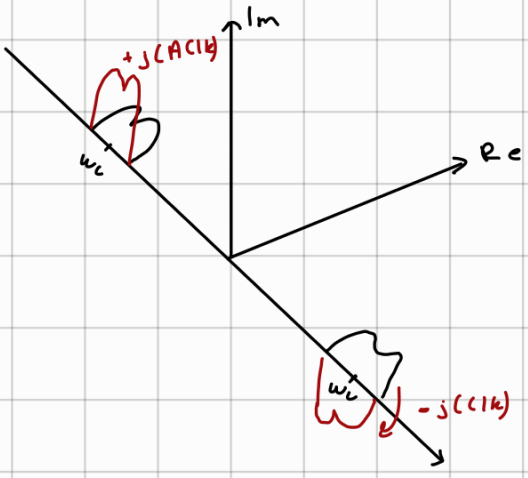
90° Phase Shift

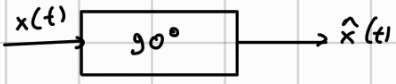
$$A(t)\cos(\omega_c t+\phi(t)+90^\circ) = A(t)\left\{\frac{e^{j[\omega_c t+\phi(t)-90^\circ]}}{2}+\frac{e^{-j[\omega_c t+\phi(t)-90^\circ]}}{2}\right\}$$

$$= A(t)\left\{-j\underbrace{\frac{e^{j[\omega_c t+\phi(t)]}}{2}}_{\text{positive freq}}+j\underbrace{\frac{e^{-j[\omega_c t+\phi(t)]}}{2}}_{\text{Negative freq}}\right\}$$

$$= A(t)\cdot\sin[\omega_c t+\phi(t)]$$



Hilbert Transform



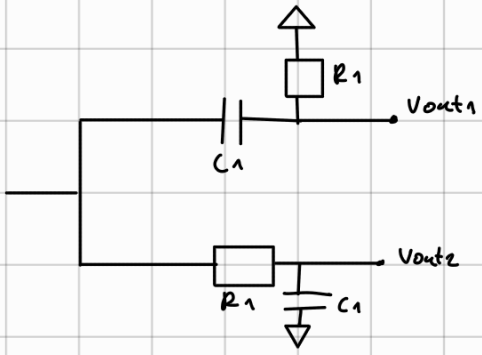
$$H(j\omega) = -j\,\mathrm{sgn}(\omega)$$

$$\mathrm{sgn}(\omega) = \begin{cases} 1 & ,\omega>0 \\ 0 & ,\omega=0 \\ -1 & ,\omega<0 \end{cases}$$

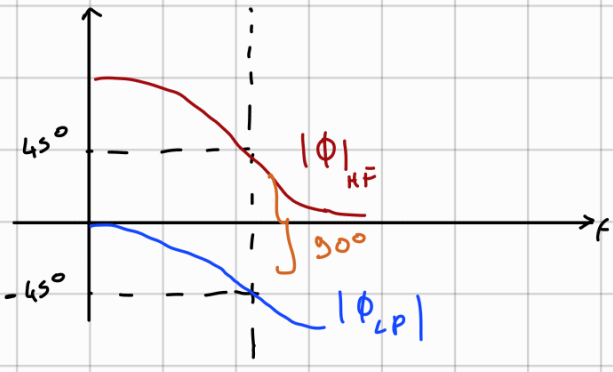
$$I(f)+jQ(f)$$

Implentation of 90° phase shift

(1) RC-CR Network

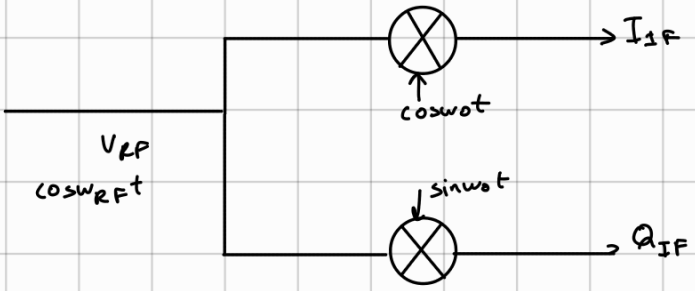




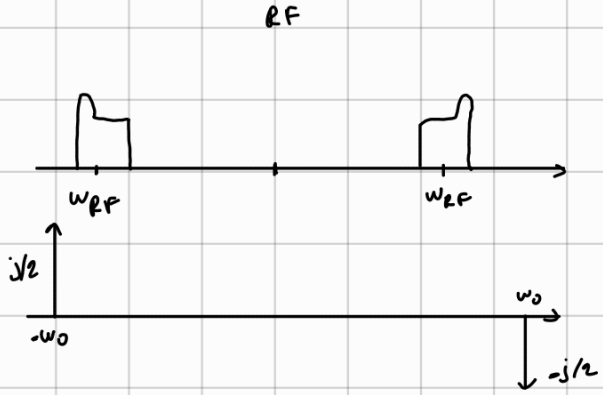


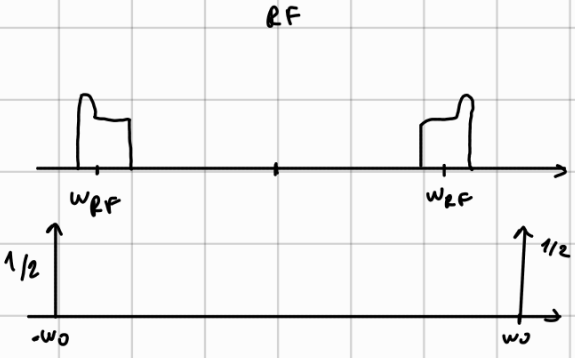
Ama sadece merkez frekansında 90 derece.

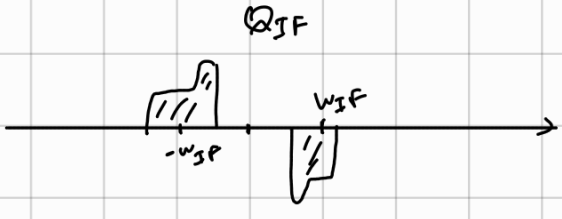
② Quadrature down Conversion



1) $w_0 > w_{RF}$ → High Injection

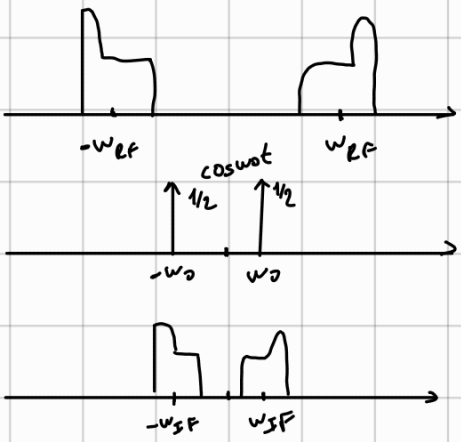


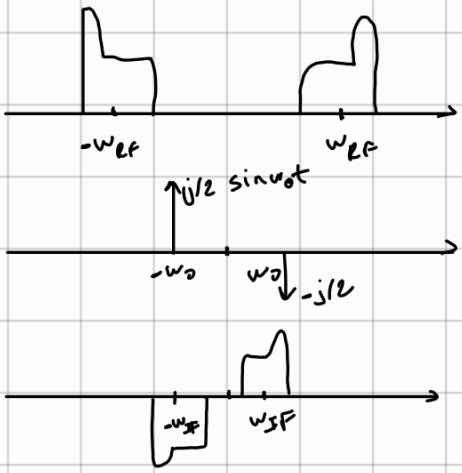






2) $w_0 < w_{RF}$ Low injection





High side

$+w_c \rightarrow$ Clk

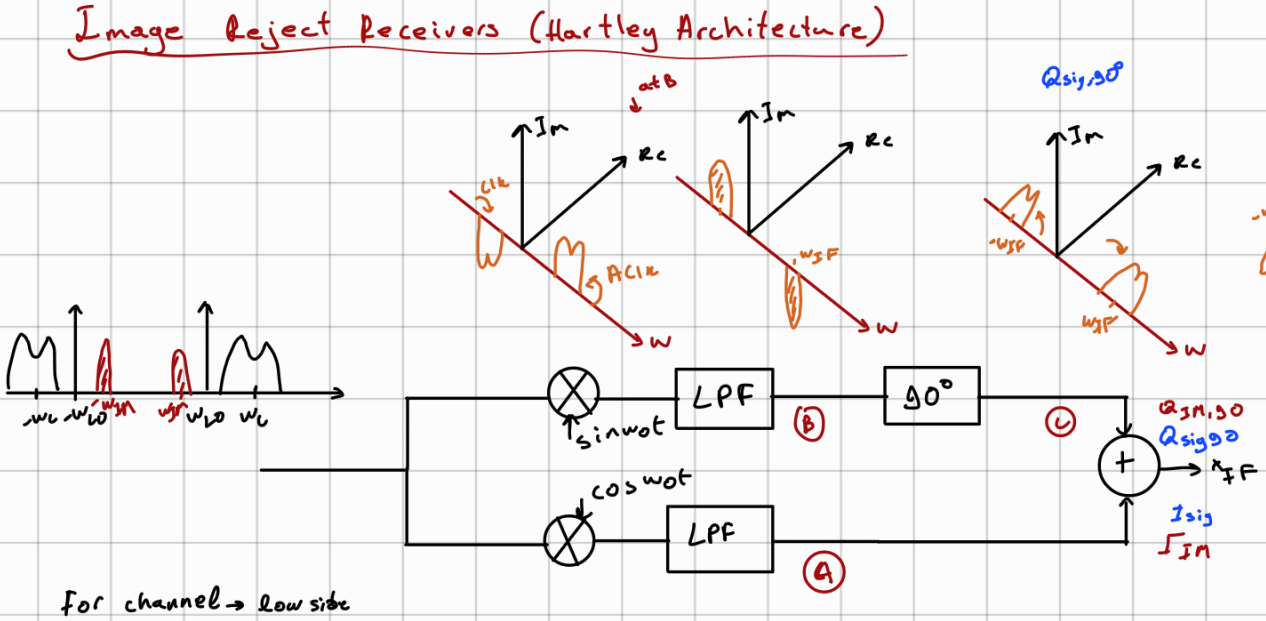
$-w_c \rightarrow$ ACLk

Low side

$+w_c \rightarrow$ ACLk

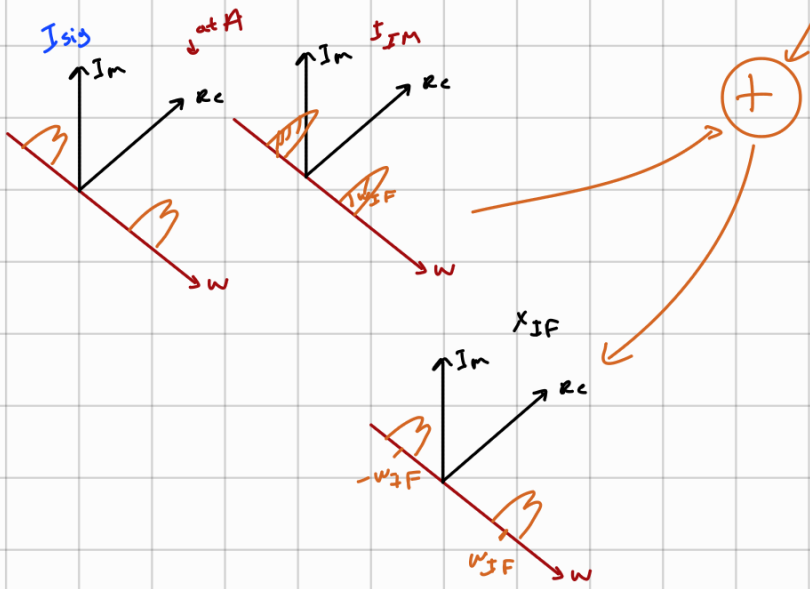
$-w_c \rightarrow$ Clk

Image Reject Receivers (Hartley Architecture)



for channel → low side

for image → high side



$$x_{in}(t) = \frac{A_{sig}}{2}\cos[\omega_{RF}+\phi_{sig}] + \frac{A_{im}}{2}\cos(\omega_{IM}+\phi_{IM})$$

$$x_A(t) = \frac{A_{sig}}{2}\cos[\underbrace{(\omega_{RF}-\omega_{LO})}_{\omega_{IF}}t+\phi_{sig}] + \frac{A_{im}}{2}\cos[\underbrace{(\omega_{im}-\omega_{LO})}_{-\omega_{IF}}t+\phi_{im}]$$

$\cos[(\omega_{LO}-\omega_{IM})t+\phi_{im}]$

$$x_B(t) = -\frac{A_{sig}}{2}\sin[\underbrace{(\omega_{RF}-\omega_{LO})}_{\omega_{IF}}t+\phi_{sig}] \pm \frac{A_{im}}{2}\sin[\underbrace{(\omega_{im}-\omega_{LO})}_{-\omega_{IF}}t+\phi_{IM}]$$

$\sin[(\omega_{LO}-\omega_{im})t+\phi_{IM}]$

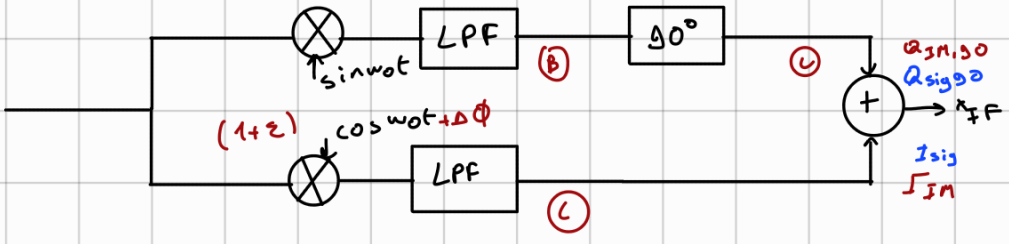
90° phase shift

$$x_C(t) = +\frac{A_{sig}}{2}\cos[(\omega_{RF}-\omega_{LO})t+\phi_{sig}] - \frac{A_{im}}{2}\cos[(\omega_{LO}-\omega_{im})t+\phi_{im}]$$

$$x_A(t)+x_B(t) = A_{sig}\cos[\omega_{IF}t+\phi_{sig}]$$

Viewer Architecture

Galiqna sorusu



$$x_{sig,out}(t) = \frac{A_{sig}}{2}\cos[(\omega_{RF}-\omega_{LO})t+\phi_{sig}] + \frac{A_{sig}}{2}(1+\varepsilon)\cos[(\omega_{RF}-\omega_{LO})t+\phi_{sig}+\Delta\theta]$$

$$x_{im,out}(t) = \frac{A_{im}}{2}(1+\varepsilon)\cos[(\omega_{IM}-\omega_{LO})t+\phi_{im}+\Delta\theta] + \frac{A_{im}}{2}\cos[(\omega_{LO}-\omega_{IM})t+\phi_{IM}]$$

In case of phase and gain mismatch (error)

Image can not remove completely

Measure of Image Rejection

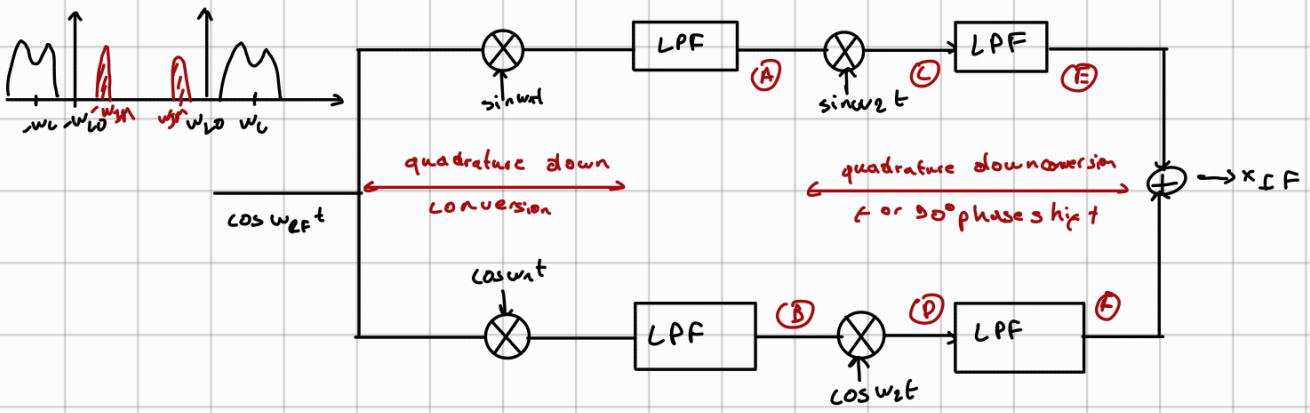
IRR : Image Rejection Ratio

$$\left.\frac{P_{im}}{P_{sig}}\right|_{out} = \frac{A_{im}^2}{A_{sig}^2}\,\frac{(1+\varepsilon)^2-2(1+\varepsilon)\cos\Delta\theta+1}{(1+\varepsilon)^2+2(1+\varepsilon)\cdot\cos\Delta\theta+1}$$

$A_{im} = A_{sig} \rightarrow$

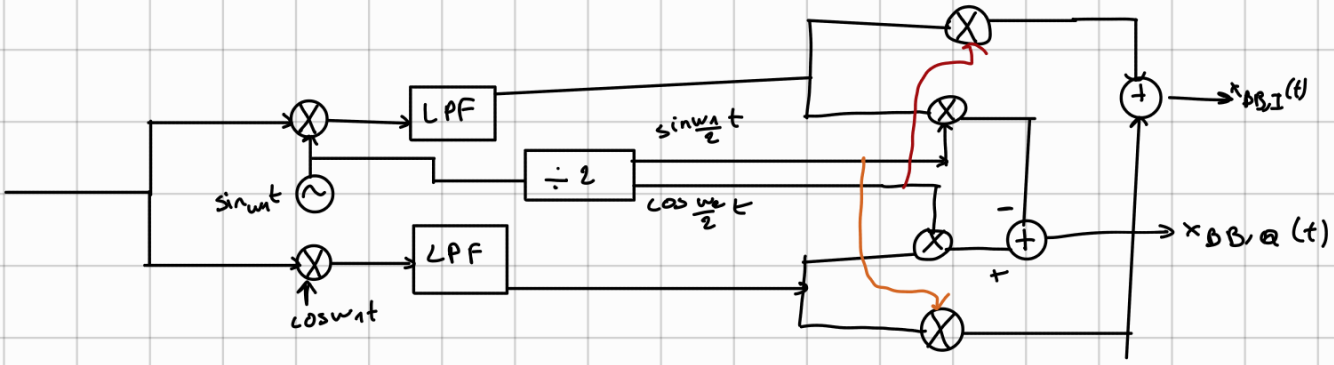
for $\Delta\theta \ll 1$, $\varepsilon \ll 1$ $\rightarrow$ $IRR = \frac{4}{\varepsilon^2+\Delta\theta^2}$

Weaver Architecture for Image Rejection



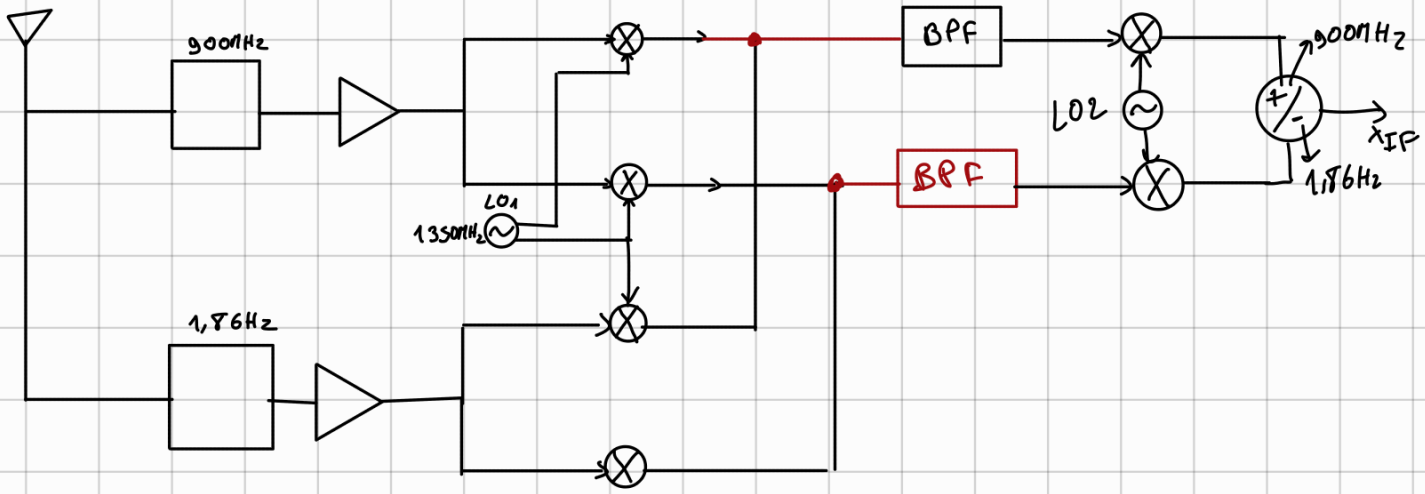
ikinci image den kurtulmak için son kısmı zero if kullanılabilir

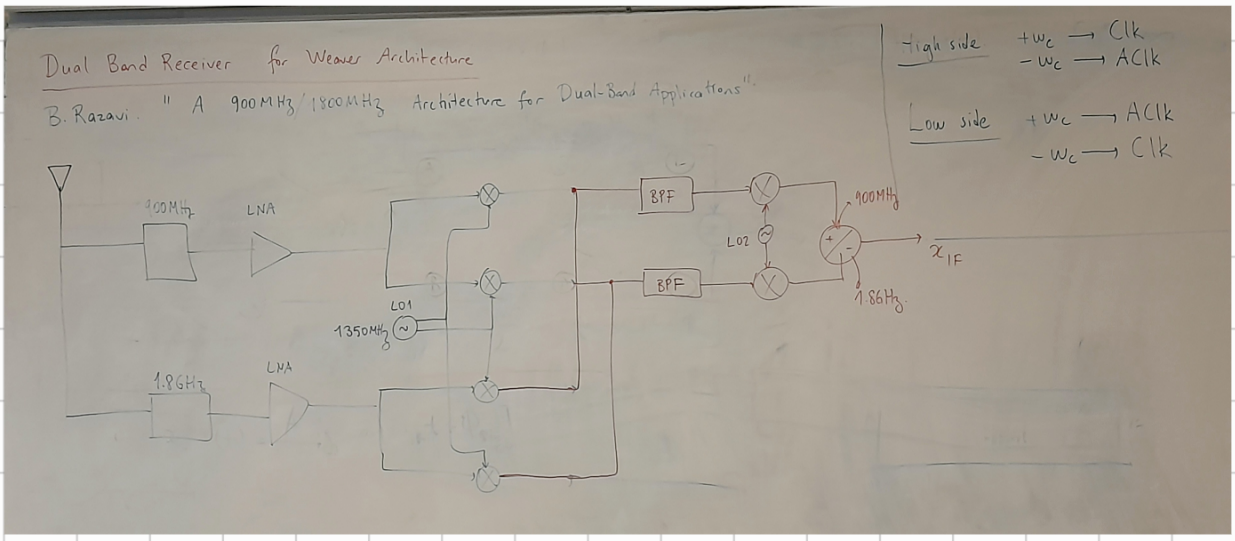
Double quadrature Down Conversion



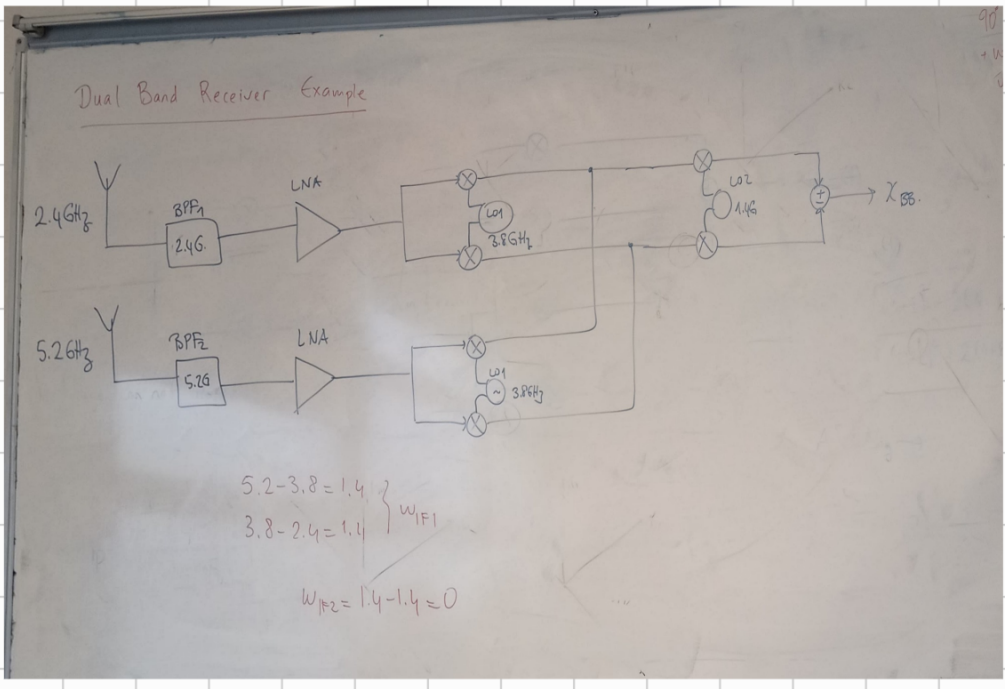
Dual Band Receiver for Weaver Architecture

B. razavi "A 900MHz/1800MHz Architecture for Dual Band Applications"





Dual Band Receiver Example



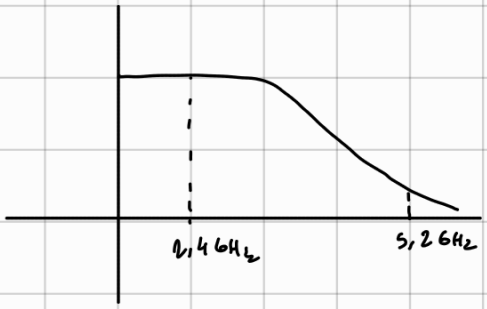
$5,2 - 3,8 = 1,4$

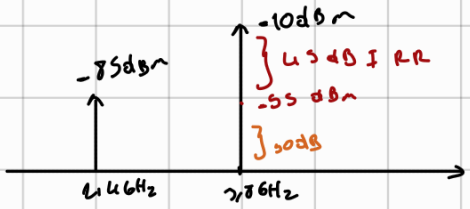
$3,8 - 2,4 = 1,4$ $\Big\}\ \omega_{IF1}$

$\omega_{IF_2} = 1,4 - 1,4 = 0$

Soru We have signal of $-85dBm$ at $2,4\,GHz$ and $-10dBm$ at $5,2\,GHz$ as interfere

What is the amount of rejection in BPF_1 at $5,2\,GHz$ for $SNR = 20dB$ & $IRR = 45dB$





$$SNR|_{dB} = S_{dBm} - N_{dBm} \geq 20dB$$

$$-85 - N_{dBm} \geq 20dB$$

$$N_{dBm} \leq -105dBm$$

$$S_{int}|_{dBm} - |BPF|_{dB} - |IRR|_{dB} = N_{dBm} \leq -105\,dBm$$

$$-10dBm - |BPF_1|_{dB} - 45 \leq -105dBm$$

$$|BPF_1| \geq 50dB$$